

Outset · AI in Healthcare · Day 3 · Group 2

Never miss a melanoma

Screening skin spots for cancer, where catching every melanoma matters more than raw accuracy

Jinchi Wei

Outset

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BACKGROUND

Two mistakes, and only one of them can be fatal

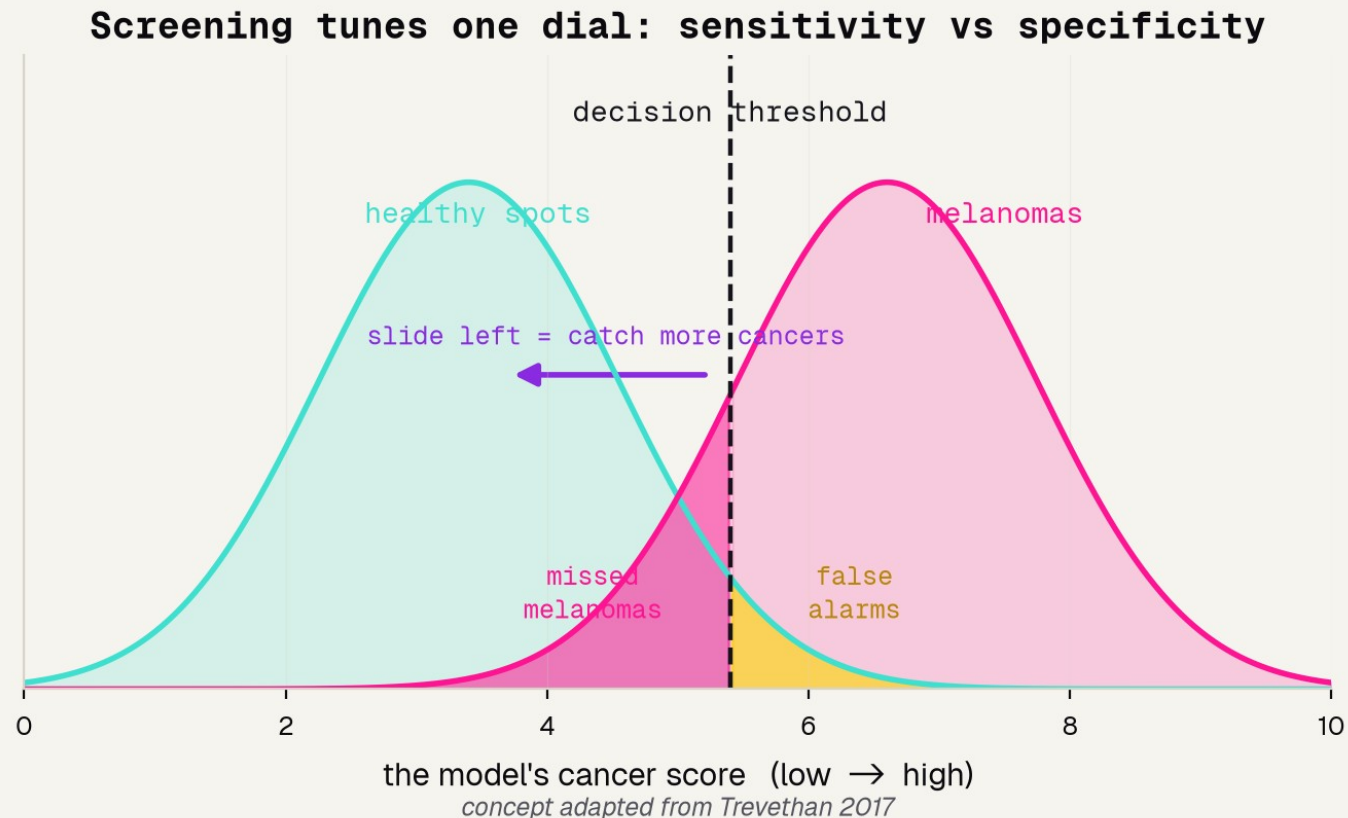
The two mistakes are not equally bad

	says: not	says: melanoma
truly melanoma	MISSED CANCER the deadly mistake	caught it correct -- refer on
truly harmless	all clear correct	false alarm just a check-up visit

adapted from Esteva 2017; Haenssle 2018

Dermatology-AI looks at a photo of a skin spot and estimates how likely it is to be cancer, and landmark studies showed a neural network can match or beat dermatologists at exactly this task (Esteva 2017, Haenssle 2018). But for a screening tool the goal is not to be right on average -- it is to never wave a real melanoma through as harmless, because that missed cancer is the one mistake that can kill.

Screening tunes one dial: sensitivity versus specificity



Every screening model has a decision threshold, and where you set it is a trade-off you cannot escape. Slide it one way and you catch more cancers but raise more false alarms; slide it the other way and you calm the false alarms but start missing real disease. Screening deliberately favors sensitivity -- catching the cancers -- because a false alarm just costs a follow-up visit (Trevethan 2017).

We borrow a brain instead of starting blank

Borrow a brain: transfer learning



The reused knowledge is why a model can learn skin cancer from thousands, not millions, of photos

adapted from Kim et al. 2022

We only have about ten thousand skin images, far too few to teach a network vision from scratch. So we use transfer learning: start from a network already trained on millions of everyday photos, keep everything it learned about edges, colors, and textures, and fine-tune only a small new head on the skin images (Kim 2022). That borrowed knowledge is why a model can learn skin cancer from thousands, not millions, of pictures.

METHODS

The task: seven kinds of skin spot, one that we cannot miss

Our data is DermaMNIST, a tidy version of the HAM10000 dermatoscopy collection with seven kinds of skin lesion, one of which is melanoma -- the class we cannot afford to miss (Tschandl 2018).

THE CATCH -- IMBALANCE

About two-thirds of every set is harmless moles. Melanoma is only a thin slice -- exactly the setup where a lazy model can look accurate while catching almost no cancers.

How we built it, and how we graded it honestly

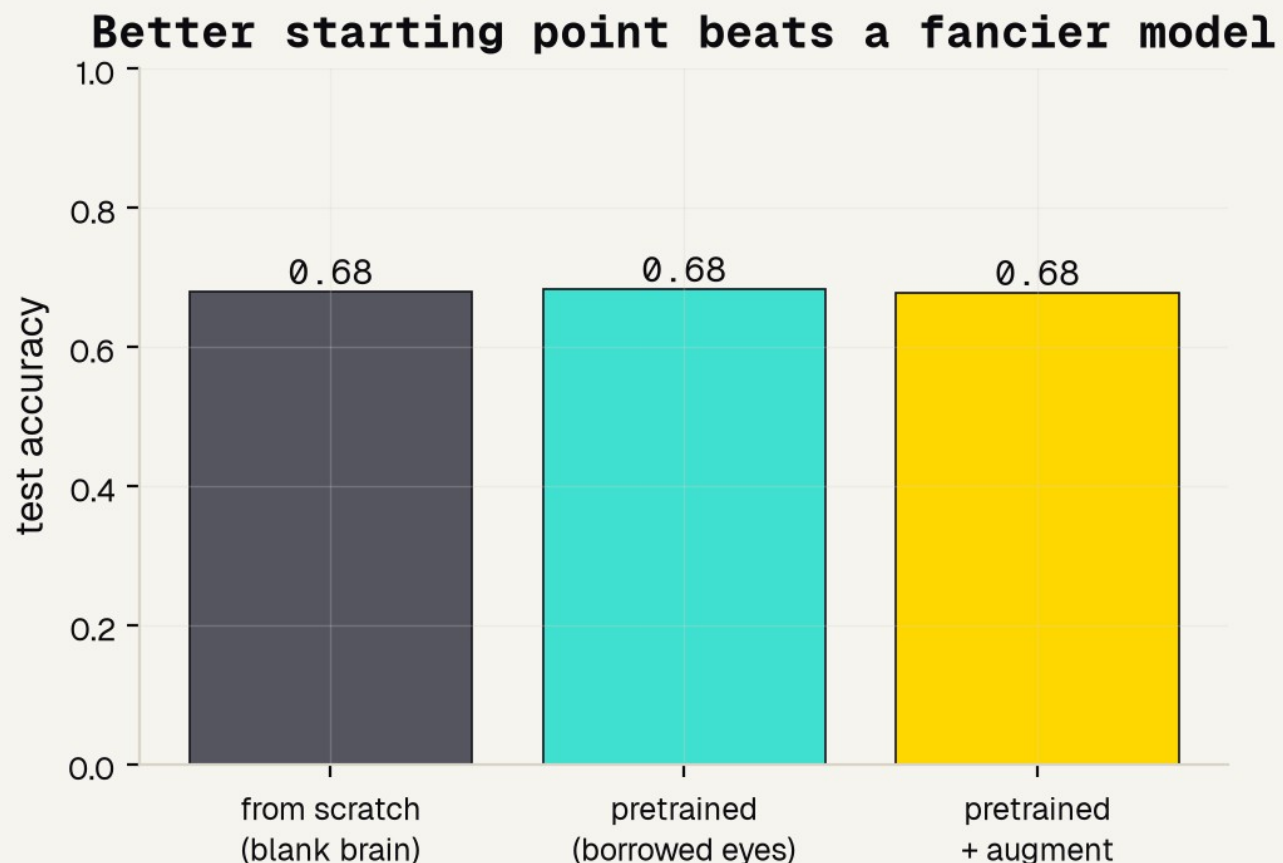
We trained the network three ways -- from scratch, pretrained, and pretrained with augmentation -- and then audited the best one the way a regulator would, refusing to grade on accuracy alone.

WE GRADED ON

Melanoma recall -- the share of true melanomas actually caught. On this data that single number, not overall accuracy, is what decides whether the tool is safe.

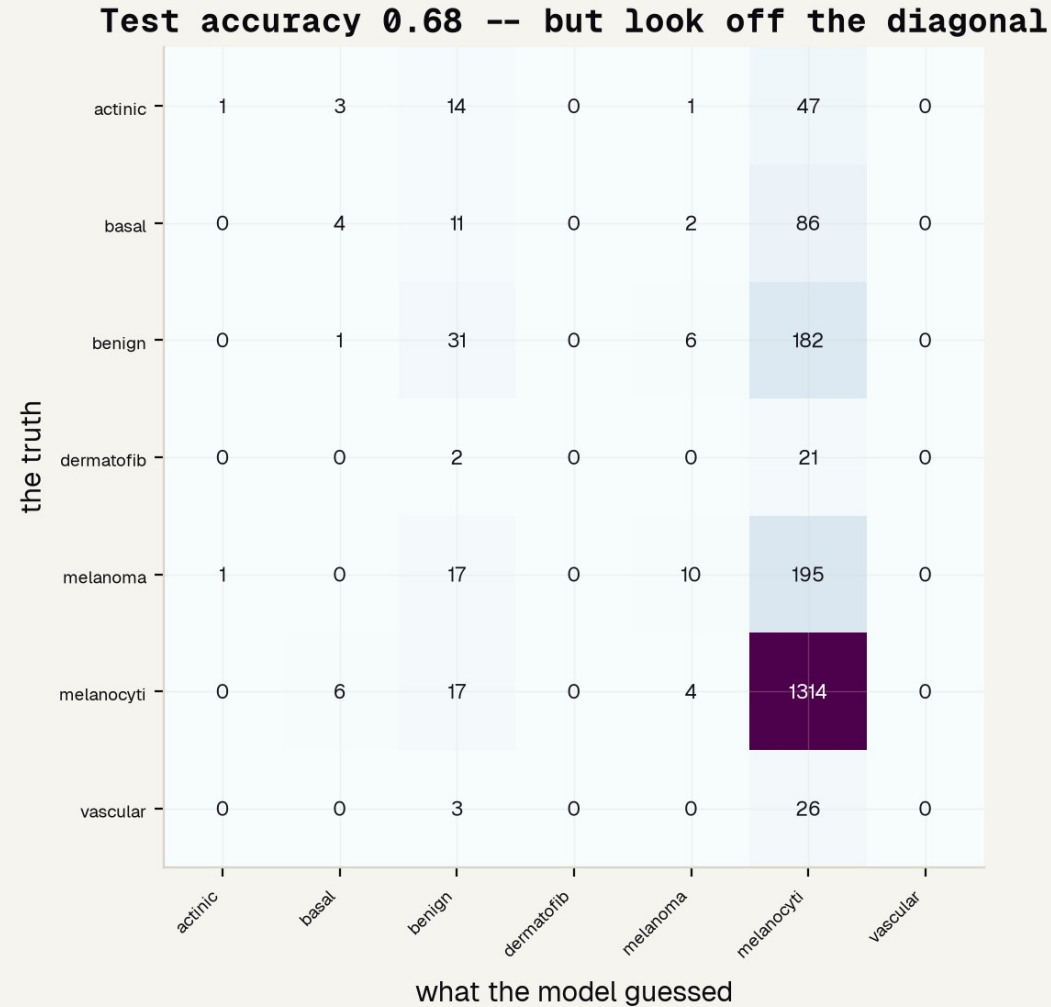
RESULTS

A better starting point beat a fancier model



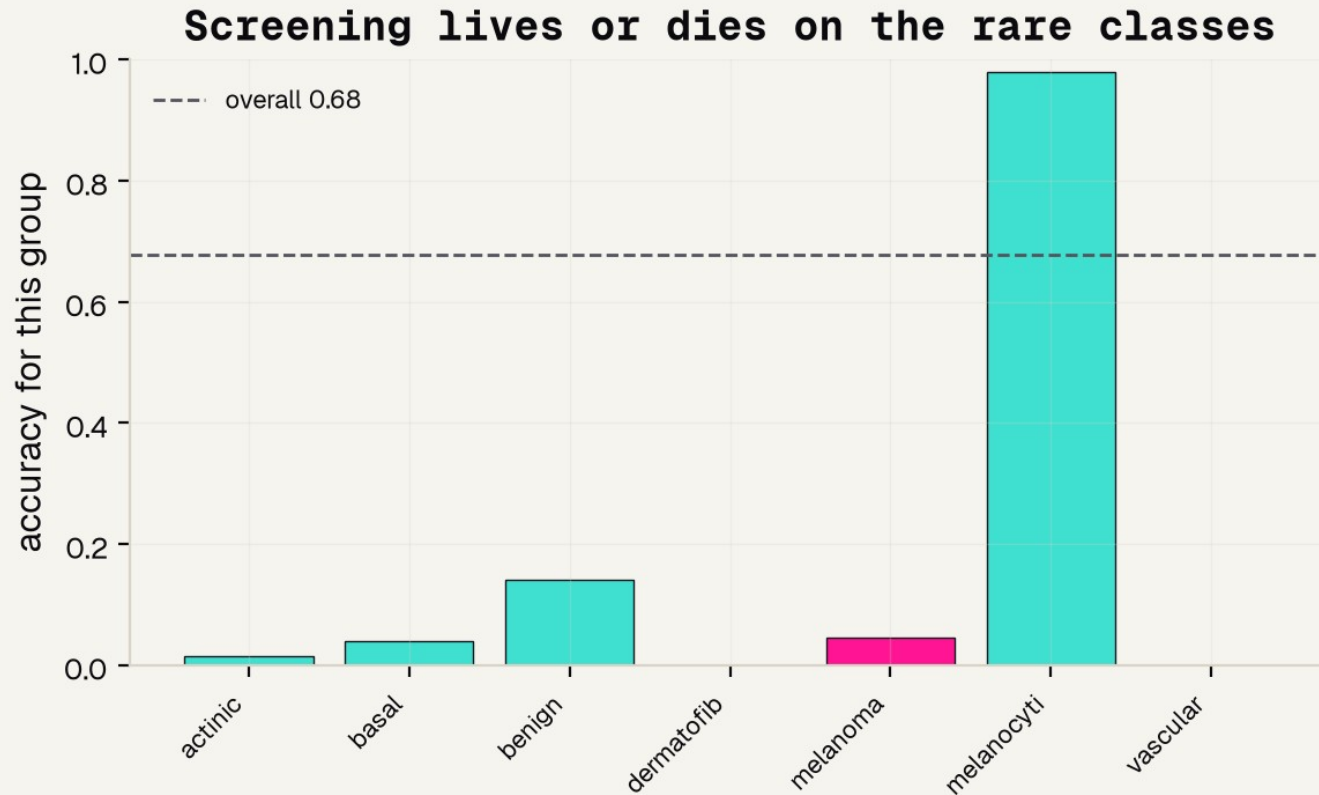
The three training setups landed almost on top of each other, and that is the honest, humbling lesson: on a dataset this small the ceiling is set by the data, not by how clever the model is. A better starting point (transfer learning) matters more than any fancy trick you bolt on afterward.

The confusion matrix tells the real story



Overall accuracy is one number; the confusion matrix shows which mistakes the model actually makes. The diagonal is correct and everything off it is an error, and one column swallows almost everything -- the model funnels nearly every spot, cancer or not, into the common harmless-mole class.

Screening lives or dies on the rare classes



Break accuracy down class by class and the average falls apart. The model is near-perfect on the one common class and near the floor on every rare one, including melanoma, highlighted in pink. A tool that works this differently across classes is not one you would ever deploy to screen a patient.

The one number that matters: melanoma recall

MELANOMA RECALL

4.5%

only 10 of 223 melanomas caught

CANCERS MISSED

213

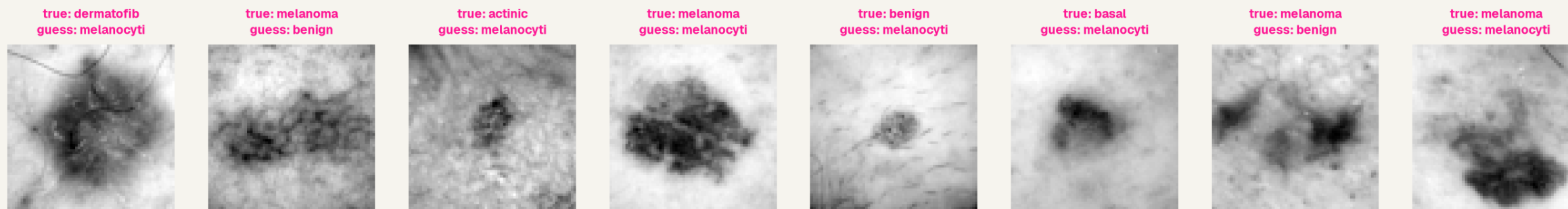
waved through as harmless

Sixty-eight percent overall accuracy hid a tool that missed almost every cancer. A healthy-looking number can conceal a screening tool that is dangerous.

The cases it got wrong are the ones that matter

The most revealing slide in any talk is the model's mistakes. Look how many true melanomas it labeled as harmless moles: every one is a false negative, the exact error a screening tool exists to prevent, and the reason a dermatologist would want a second look at all of them.

The cases it got wrong



BEING HONEST

A toy model, and why that is exactly the point

This model is far too small and simple to trust on a real patient: sixty-four-pixel grayscale thumbnails throw away the color and fine texture that dermatologists and real systems rely on, and its melanoma recall is nowhere near safe (Wei 2024). The point was never the score -- it was learning the screening mindset: never miss the dangerous case, and never let a shiny number fool you.

A REAL TOOL WOULD

Train on full-resolution color images. Deliberately tune its threshold toward high sensitivity. Be validated across many hospitals before it ever sees a patient.

References, and thank you

SEVEN PAPERS GROUNDED THIS PROJECT

[1] Esteva et al. 2017 -- Dermatologist-level classification of skin cancer with deep neural networks. Nature.

[2] Tschandl et al. 2018 -- The HAM10000 dermatoscopy dataset. Scientific Data.

[3] Haenssle et al. 2018 -- Man against machine: a CNN vs 58 dermatologists. Annals of Oncology.

[4] Tschandl et al. 2019 -- Human readers vs machine learning for pigmented lesions. Lancet Oncology.

[5] Kim et al. 2022 -- Transfer learning for medical image classification: a review. BMC Medical Imaging.

[6] Trevethan 2017 -- Sensitivity, specificity, and predictive values. Frontiers in Public Health.

[7] Wei et al. 2024 -- Artificial intelligence and skin cancer. Frontiers in Medicine.

Thank you for watching -- go build something you can honestly evaluate.

Thank You

魏晉祺

Jinchi Wei

Outset

jinchikwei@gmail.com

夢想